# IETF-125 Hackathon

I2ICF Side Meeting

# 5G-I2NSF: An Integrated Security System for 5G Networks with I2NSF

March 15, 2026, Shenzhen

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Members: Joseph Ahn, Jiwon Suh, and Jiwon Yoo, and Bien Aime Mugabarigira

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# IETF-125 5G-I2NSF System for Integrated Security Services in 5G Networks


## Champion: Jaehoon Paul Jeong (SKKU)

## IETF-125 5G-I2NSF Project

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- Yong-Geun Hong (DJU)
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### Researchers:

- Jung-Soo Park (ETRI)
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- Jiwon Suh (SKKU)
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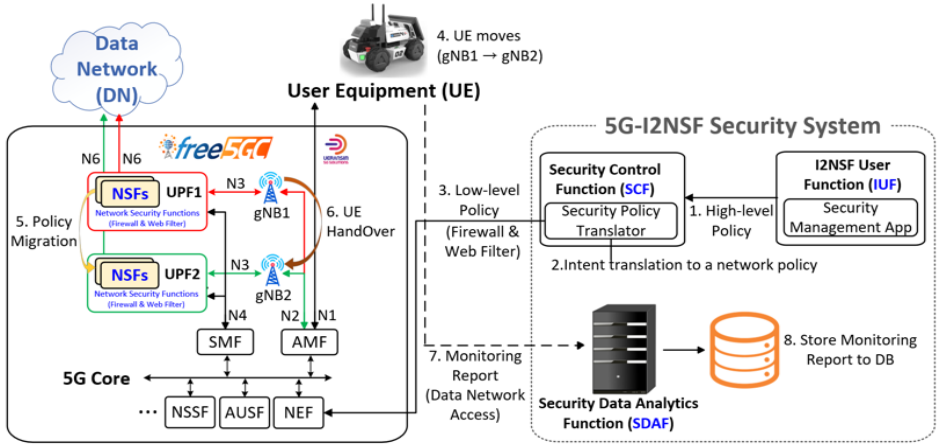
## 5G-I2NSF System




## Objectives

- This project proposes an integrated framework for automated security orchestration in 5G edge environments based on the Interface to Network Security Functions (I2NSF) framework.
- This framework translates high-level YANG/XML security policies into low-level rules and dynamically deploys Network Security Functions (NSFs) within a User Plane Function (UPF).
- It supports seamless security policy migration during a User Equipment (UE)'s handover to maintain consistent security enforcement with minimal latency.

## Future Work

- We will employ a Large Language Model (LLM) to interpret natural language intents and automatically generate YAML-based policies.
- AI-driven analysis will further enhance security services through adaptive learning, anomaly detection, and policy refinement.

## 5G-I2NSF Development Environment

- OS: Ubuntu 22.04
- Free5GC VM: version 4.1.0
- UERANSIM VM (UE & RAN): version 3.2.6
- GitHub Repository: https://github.com/jaehoonpauljeong/5G-I2NSF/tree/main/IETF-125

## Workflow of the 5G-I2NSF Testbed

1. A high-level security policy is first created by an I2NSF User Function (IUF) according to a user's intent, and a Security Control Function (SCF) converts it into a corresponding low-level security policy.
2. This policy is forwarded to a Session Management Function (SMF) through a Network Exposure Function (NEF), enabling the required Network Security Functions (NSFs) to be instantiated in a UPF.
3. When a UE moves from a base station (gNB1) to another base station (gNB2) through handover, the 5G system anticipates the next serving gNB and proactively transfers the NSF's security policies (e.g., firewall and web filtering) from the current UPF (UPF1) to the next UPF (UPF2).
4. An Access and Mobility Management Function (AMF) coordinates the handover of the moving UE while the SMF reestablishes the N4 session with UPF2.
5. The UE's traffic logs are sent to a Security Data Analytics Function (SDAF) as an Analyzer to verify whether the deployed NSFs properly enforce the security policies or not.

# Goal of Hackathon Project

- **Goal**
  - To make an Edge-Based Security System in 5G Networks with the Interface to Network Security Functions (I2NSF) Framework which is the IETF standard.

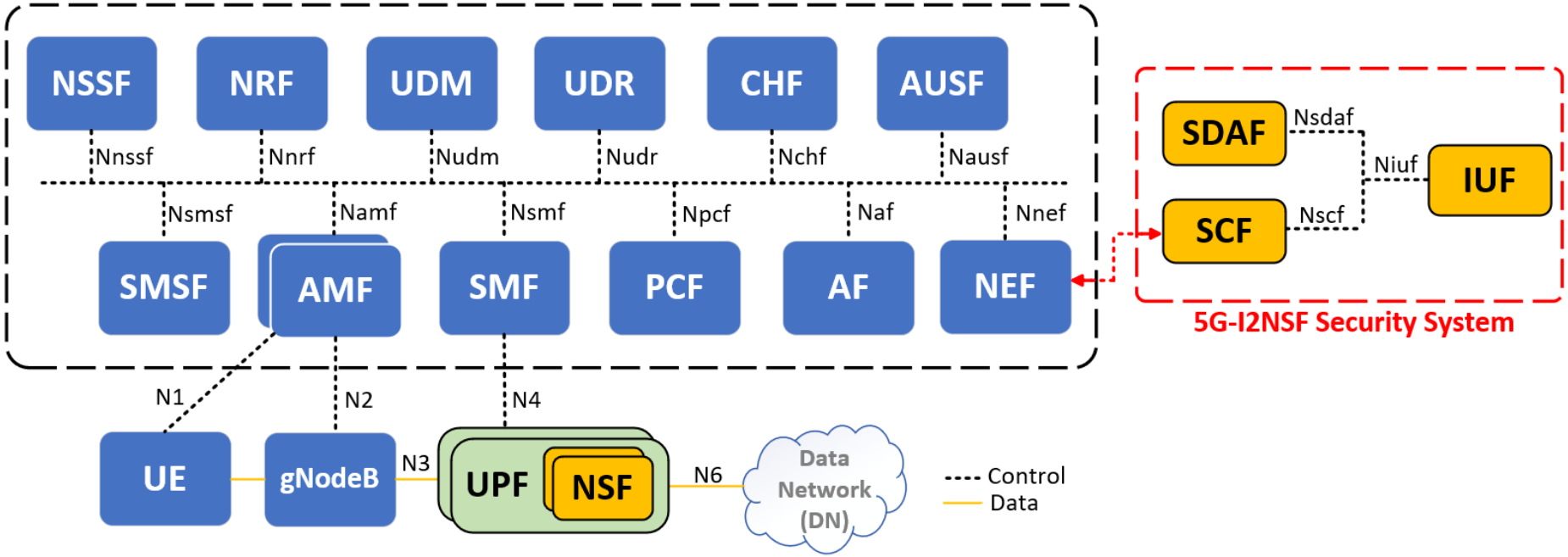
- **Security Policy Provisioning** through **Edge-Based I2NSF Framework**
  - Integration of I2NSF to 5G as Edge Approach rather than Cloud Approach.
  - Formation of I2NSF Components as 5G Network Functions (NFs) and also I2NSF Interfaces as 5G Interfaces.

- **Internet Drafts for the 5G-I2NSF Project**
  - https://datatracker.ietf.org/doc/draft-ahn-nmrg-5g-security-i2nsf-framework/

# I2NSF-based Integration for 5G Security Services

## ➢ Integration of I2NSF-based Components in 5G Service-Based Architecture

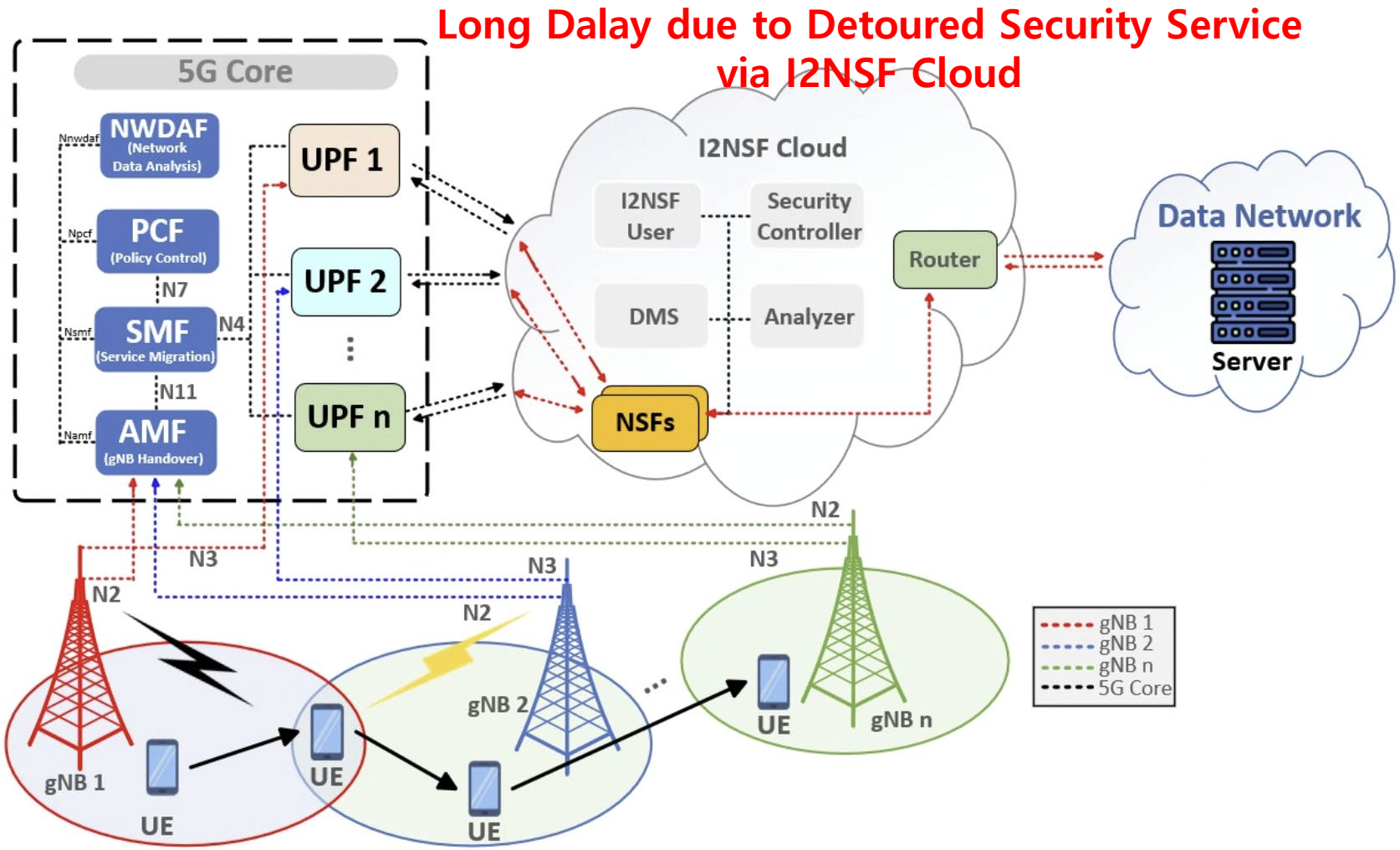
- To enhance edge-based security management in 5G networks, the I2NSF framework with I2NSF components (e.g., IUF, SCF, and SDAF) is integrated with the 5G Core Network.
  - IUF (I2NSF User Function), SCF (Security Control Function), and SDAF (Security Data Analytics Function)
- The I2NSF framework generates security policies, enforces them, and analyzes security activities through NSFs (Network Security Functions) deployed with UPFs for efficient security.




**An Integrated Security System for 5G Networks with I2NSF**

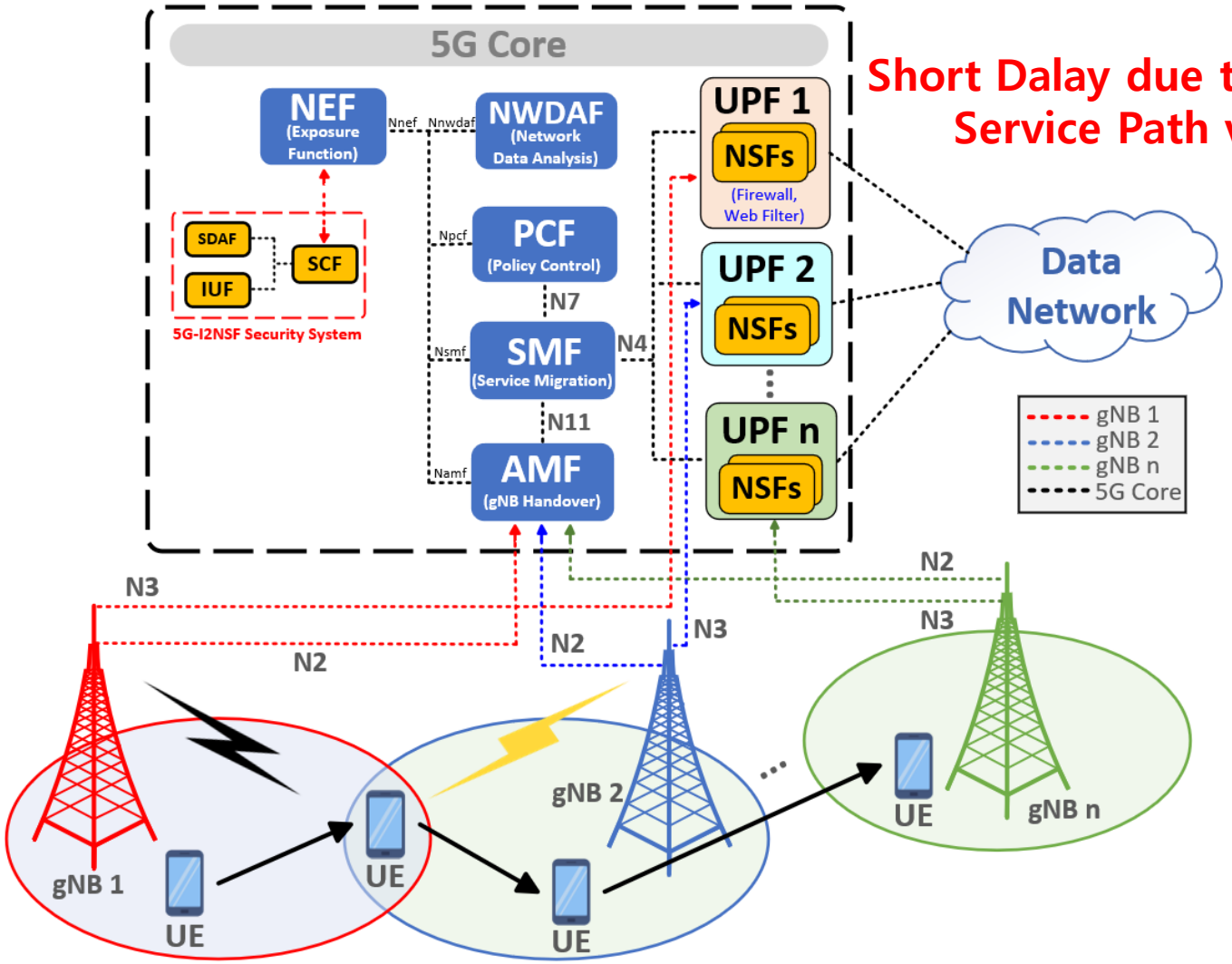
# I2NSF Cloud Approach for 5G Networks

- **Legacy:** Execution of NSFs in I2NSF Cloud outside of 5G Core Networks
  - Long delay by a detoured path from a UE to I2NSF Cloud for security services.



# I2NSF Edge Approach for 5G Networks

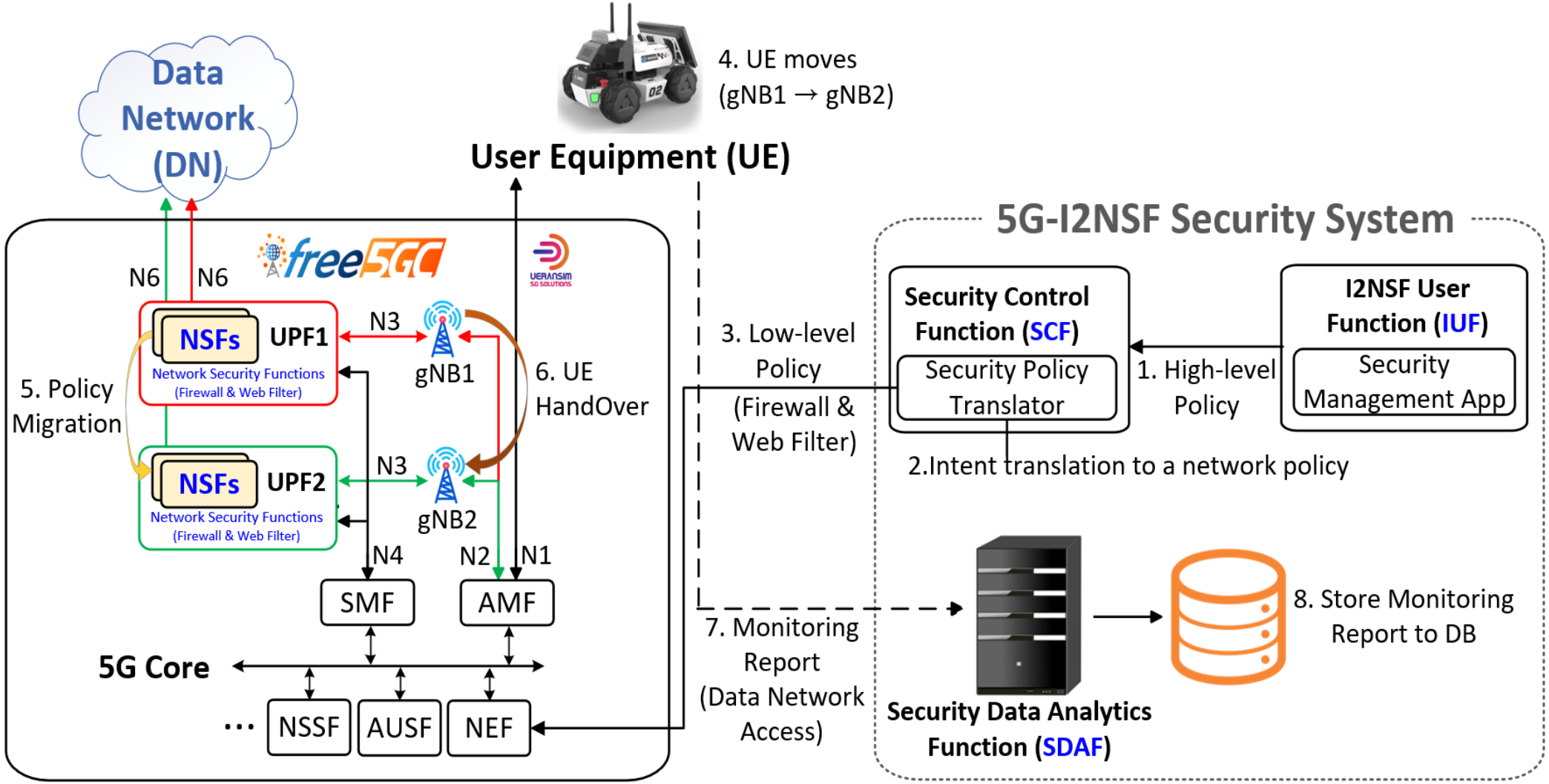
- **Legacy:** Execution of NSFs in I2NSF Cloud inside of 5G Core Networks
  - Short delay by an optimal path from a UE to I2NSF Edge for security services.




**Short Dalay due to Optimal Security Service Path via I2NSF Edge**

# An Integrated Security System for 5G Networks with I2NSF

## ➢ 5G-I2NSF System



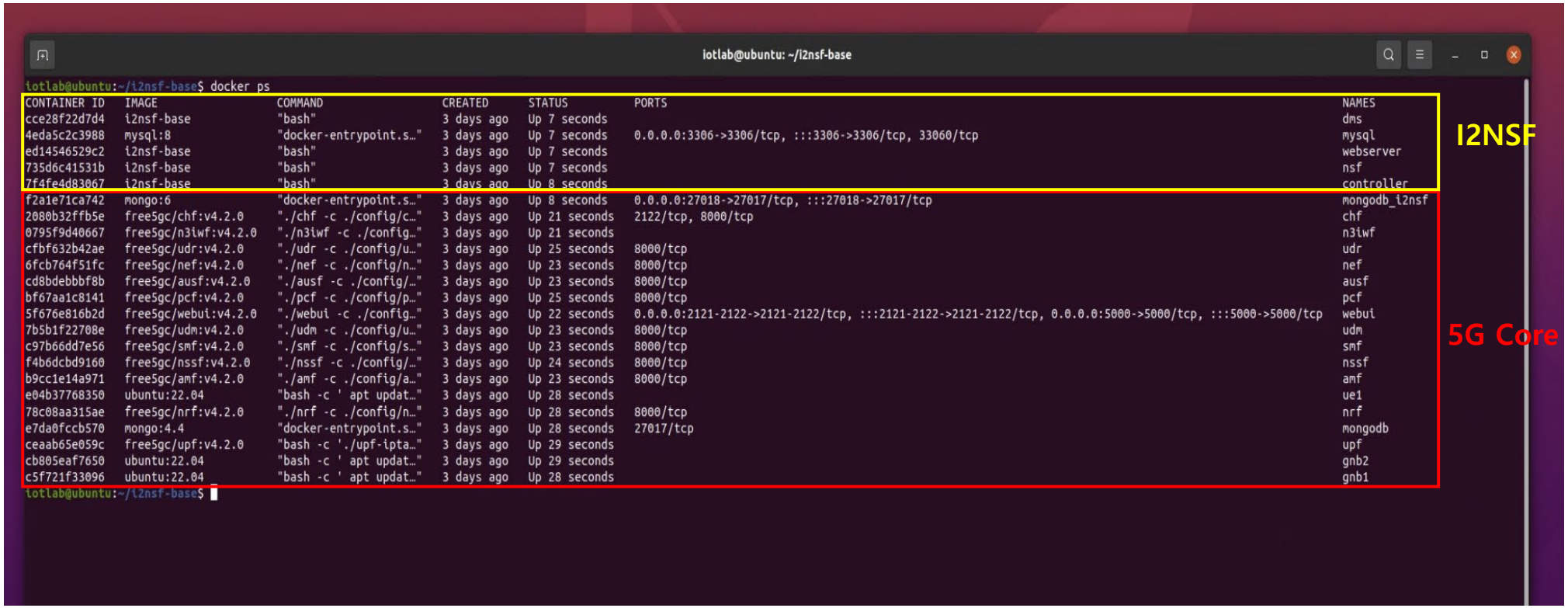
## What we learned

- We designed and deployed the 5G-I2NSF system in a containerized environment using technologies such as Docker.

- This allowed the entire testbed system to run in a containerized environment while integrating I2NSF security functions with the Free5GC 5G network.
  - I2NSF and Free5GC are running in a Cloud-Native Environment.

# Demonstration of 5G-I2NSF (1/4)

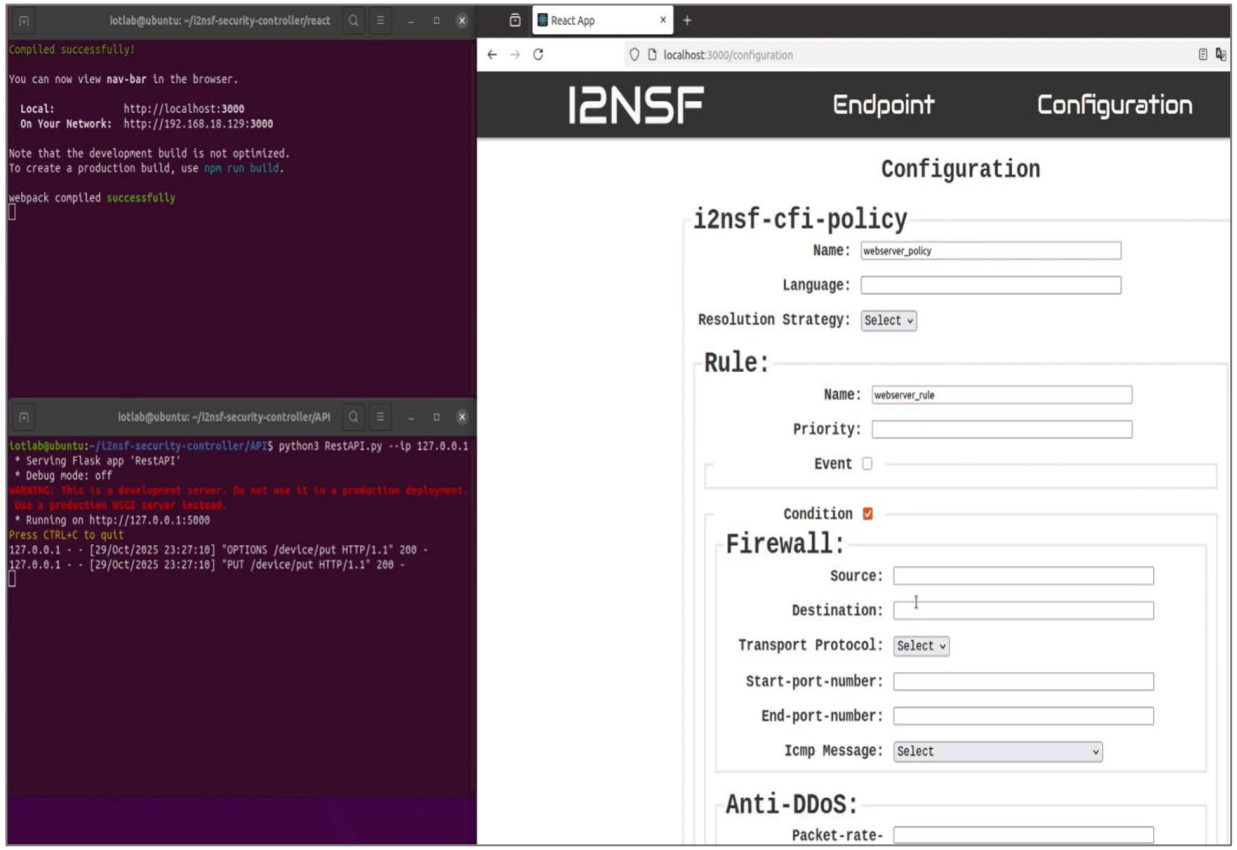
## ➢ Containerized 5G Core Network with I2NSF Framework



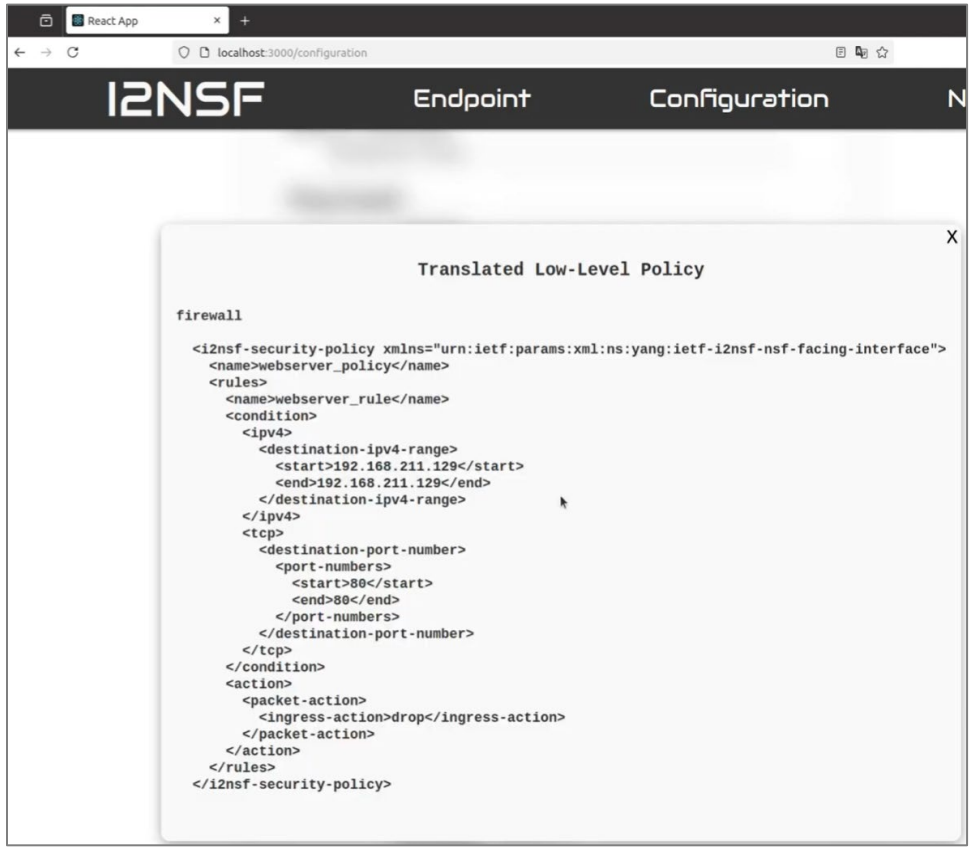

**Docker Containers for I2NSF NSFs and Free5GC 5G Core Network**

# Demonstration of 5G-I2NSF (2/4)

## ➢ Security Policy Translation




I2NSF User Function (IUF) for High-level Security Policy




Security Control Funtion (SCF) for
Low-level Security Policy Generation

# Demonstration of 5G-I2NSF (3/4)

## ➢ NSF (Firewall) Generation on UPF

```
root@ubuntu: /home/iotlab/i2nsf-nsf/firewall
e -DCONFD_C_PRODUCT_CONFD
cc notifier_builtin_replay_store.o /home/iotlab/confd/lib/libconfd.a -lpthread -lm  -Wall -g -I/home/iotlab/c
onfd/include -DCONFD_C_PRODUCT_CONFD  -o notifier_builtin_replay_store
/home/iotlab/confd/bin/confdc --fail-on-warnings  -c -o ietf-ethertypes.fxs  ietf-ethertypes.yang
/home/iotlab/confd/bin/confdc --fail-on-warnings  -c -o ietf-interfaces.fxs  ietf-interfaces.yang
/home/iotlab/confd/bin/confdc --fail-on-warnings  -c -o ietf-packet-fields.fxs  ietf-packet-fields.yang
/home/iotlab/confd/bin/confdc --fail-on-warnings  -c -o ietf-i2nsf-nsf-facing-interface.fxs  ietf-i2nsf-nsf-f
acing-interface.yang
/home/iotlab/confd/bin/confdc --fail-on-warnings  -c -o ietf-i2nsf-monitoring-interface.fxs  ietf-i2nsf-monit
oring-interface.yang
/home/iotlab/confd/bin/confdc  --emit-python ietf-i2nsf-nsf-facing-interface_ns.py ietf-i2nsf-nsf-facing-inte
rface.fxs
/home/iotlab/confd/bin/confdc  --emit-python ietf-i2nsf-monitoring-interface_ns.py ietf-i2nsf-monitoring-inte
rface.fxs
Warning: the following symbols have been suppressed due to a
conflict with an enum or bit with the same mapped name but a different value:
 'logout', 'user'.
Use tailf:code-name on the conflicting enums, bits, or nodes
to avoid the conflict.
/home/iotlab/confd/bin/confdc  --emit-python notif_ns.py notif.fxs
mkdir -p ./confd-cdb
cp /home/iotlab/confd/var/confd/cdb/aaa_init.xml ./confd-cdb
ln -s /home/iotlab/confd/etc/confd/ssh ssh-keydir
Build complete
### Stopping any confd daemon
/home/iotlab/confd/bin/confd --stop || true
killall -r notifier || true
notifier: no process found
sed 's/[ \t]*<!-- enableBuiltinReplayStore -->/          <enabled>true<\/enabled>/' confd.conf > confd_prim.c
onf
/home/iotlab/confd/bin/confd -c ./confd_prim.conf --addloadpath /home/iotlab/confd/etc/confd
python3 ./firewall.py --debuglevel t
confd port: 4565
Subscribed to /i2nsf-security-policy
policy num: 1
!!test: None
test name: webserver_policy
rule-name: webserver_rule
destination_ipv4_range: [192.168.211.129 192.168.211.129]
ingress_action: idref<414370175>
Configuration applied to firewall
```

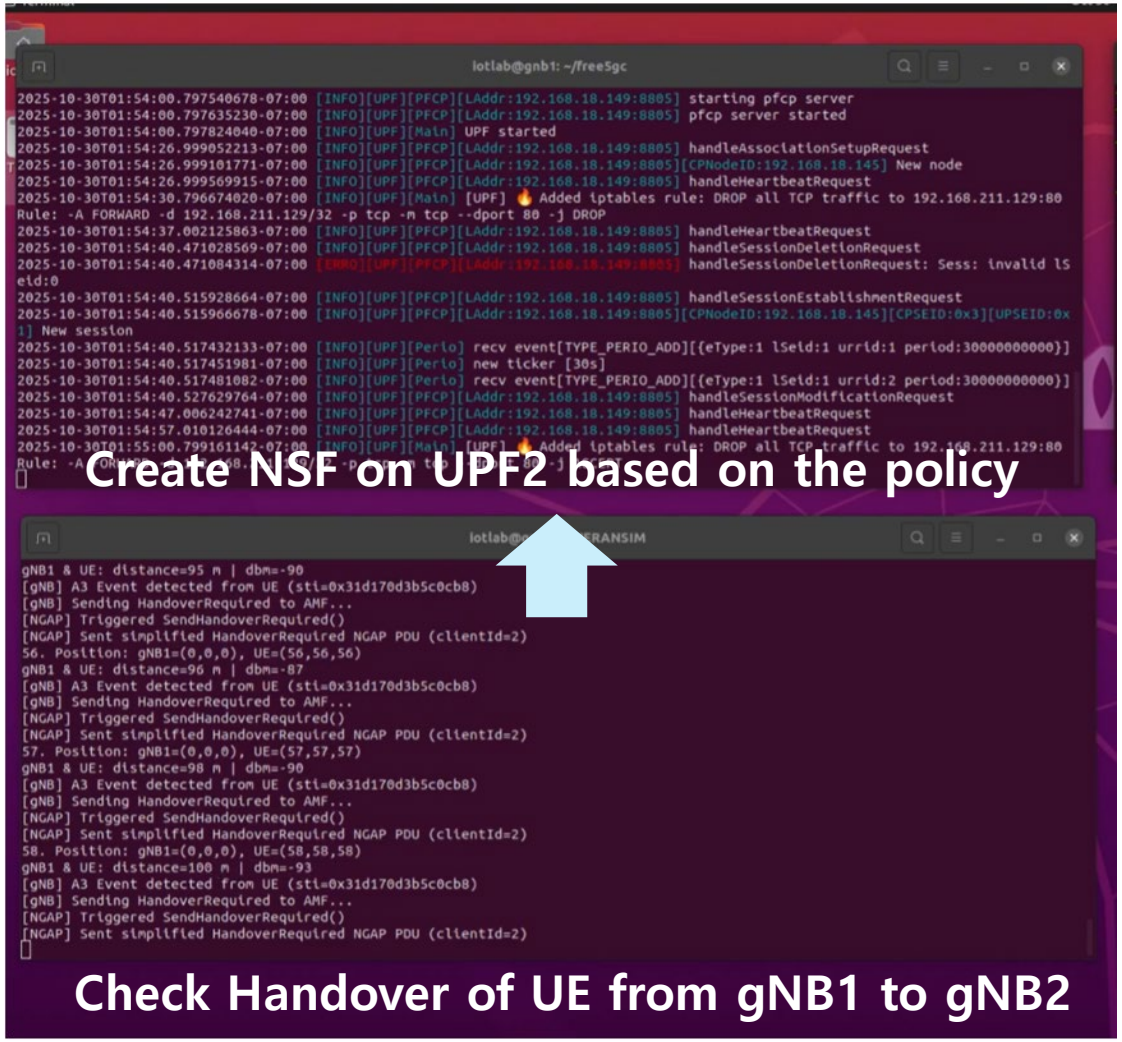
Low-level Policy application on UPF1

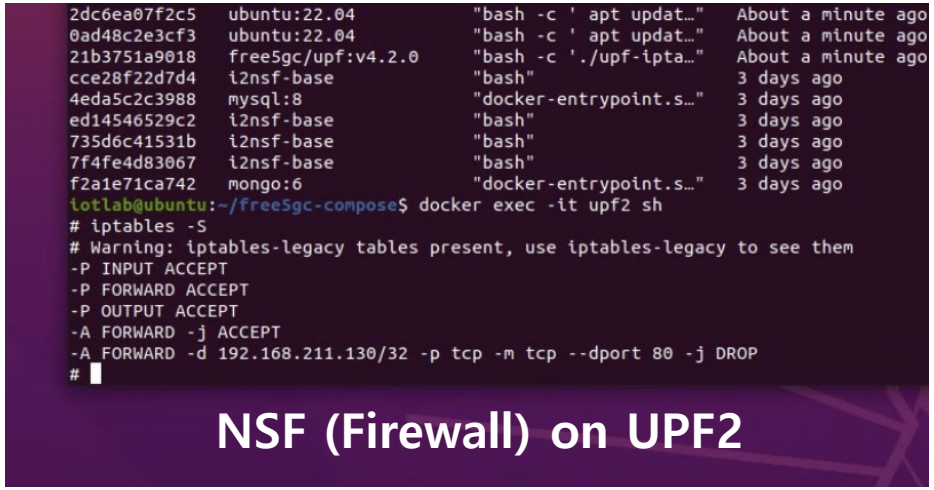
```
cfbf632b42ae   free5gc/udr:v4.2.0      "./udr -c ./config/u…"   3 days ago   Up
6fcb764f51fc   free5gc/nef:v4.2.0      "./nef -c ./config/n…"   3 days ago   Up
cd8bdebbbf8b   free5gc/ausf:v4.2.0     "./ausf -c ./config/…"   3 days ago   Up
bf67aa1c8141   free5gc/pcf:v4.2.0      "./pcf -c ./config/p…"   3 days ago   Up
5f676e816b2d   free5gc/webui:v4.2.0    "./webui -c ./config…"   3 days ago   Up
7b5b1f22708e   free5gc/udm:v4.2.0      "./udm -c ./config/u…"   3 days ago   Up
c97b66dd7e56   free5gc/smf:v4.2.0      "./smf -c ./config/s…"   3 days ago   Up
f4b6dcbd9160   free5gc/nssf:v4.2.0     "./nssf -c ./config/…"   3 days ago   Up
b9cc1e14a971   free5gc/amf:v4.2.0      "./amf -c ./config/a…"   3 days ago   Up
e04b37768350   ubuntu:22.04            "bash -c ' apt updat…"   3 days ago   Up
78c08aa315ae   free5gc/nrf:v4.2.0      "./nrf -c ./config/n…"   3 days ago   Up
e7da0fccb570   mongo:4.4               "docker-entrypoint.s…"   3 days ago   Up
ceaab65e059c   free5gc/upf:v4.2.0      "bash -c './upf-ipta…"   3 days ago   Up
cb805eaf7650   ubuntu:22.04            "bash -c ' apt updat…"   3 days ago   Up
c5f721f33096   ubuntu:22.04            "bash -c ' apt updat…"   3 days ago   Up
iotlab@ubuntu:~/i2nsf-base$ docker exec -it upf sh
# iptables -S
# Warning: iptables-legacy tables present, use iptables-legacy to see them
-P INPUT ACCEPT
-P FORWARD ACCEPT
-P OUTPUT ACCEPT
-A FORWARD -j ACCEPT
-A FORWARD -d 192.168.211.129/32 -p tcp -m tcp --dport 80 -j DROP
#
```

NSF (Firewall) on UPF1

# Demonstration of 5G-I2NSF (4/4)

➢ **UE Handover and Policy-based NSF Creation on UPF**



# Open-Source Project for 5G-I2NSF

[URL] https://github.com/jaehoonpauljeong/5G-I2NSF/tree/main/IETF-125

main | 1 Branch | 0 Tags | Go to file | Code

ahnjs124 Add files via upload — d4ff70b · 4 days ago — 18 Commits

| | | |
|---|---|---|
| IETF-124 | Add files via upload | 4 months ago |
| IETF-125 | Add files via upload | 4 days ago |
| 5G-I2NSF System.png | Add files via upload | 4 months ago |
| README.md | Rename 'Future Work' section to 'Next Step' | 4 months ago |

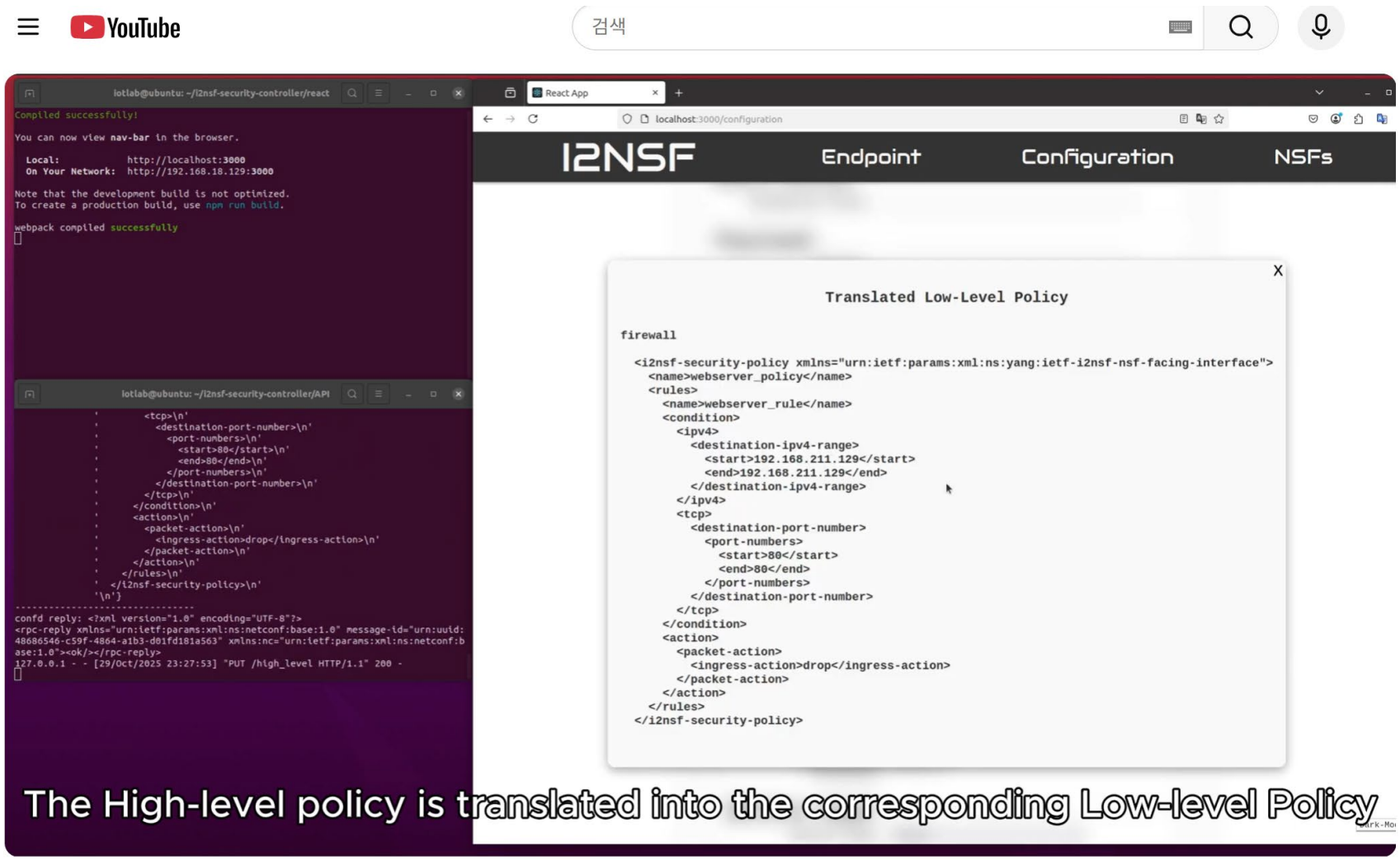
README

## 5G-I2NSF

- This study presents an integrated framework for automated security orchestration in 5G edge networks based on the Interface to Network Security Functions (I2NSF) system.
- The proposed system translates a high-level security policy in YANG/XML into the corresponding low-level security policy in YANG/XML and dynamically deploys the corresponding Network Security Functions (NSFs) within the User Plane Function (UPF).
- It also supports seamless policy migration during UE handovers, minimizing latency and ensuring consistent security enforcement across distributed edge environments.

# Demonstration Video Clip for 5G-I2NSF

[URL] https://www.youtube.com/watch?v=YxX7MGFy65E



## Next Steps

- We will design and implement the 5G protocol procedure for I2NSF considering the interaction among UE, AMF, SMF, SCF, and NSFs.
  - Under a handover scenario, a security policy will migrate from the current UPF of a UE to the next UPF of the UE.
  - An NSF will be launched proactively before the security policy migration.

- We will reflect our hackathon experience on our draft:
  - https://datatracker.ietf.org/doc/draft-ahn-nmrg-5g-security-i2nsf-framework/.

# 5G-I2NSF Hackathon Team

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- Jaehoon Paul Jeong (SKKU)
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- Yong-Geun Hong (DJU)
- Joo-Sang Youn (DEU)
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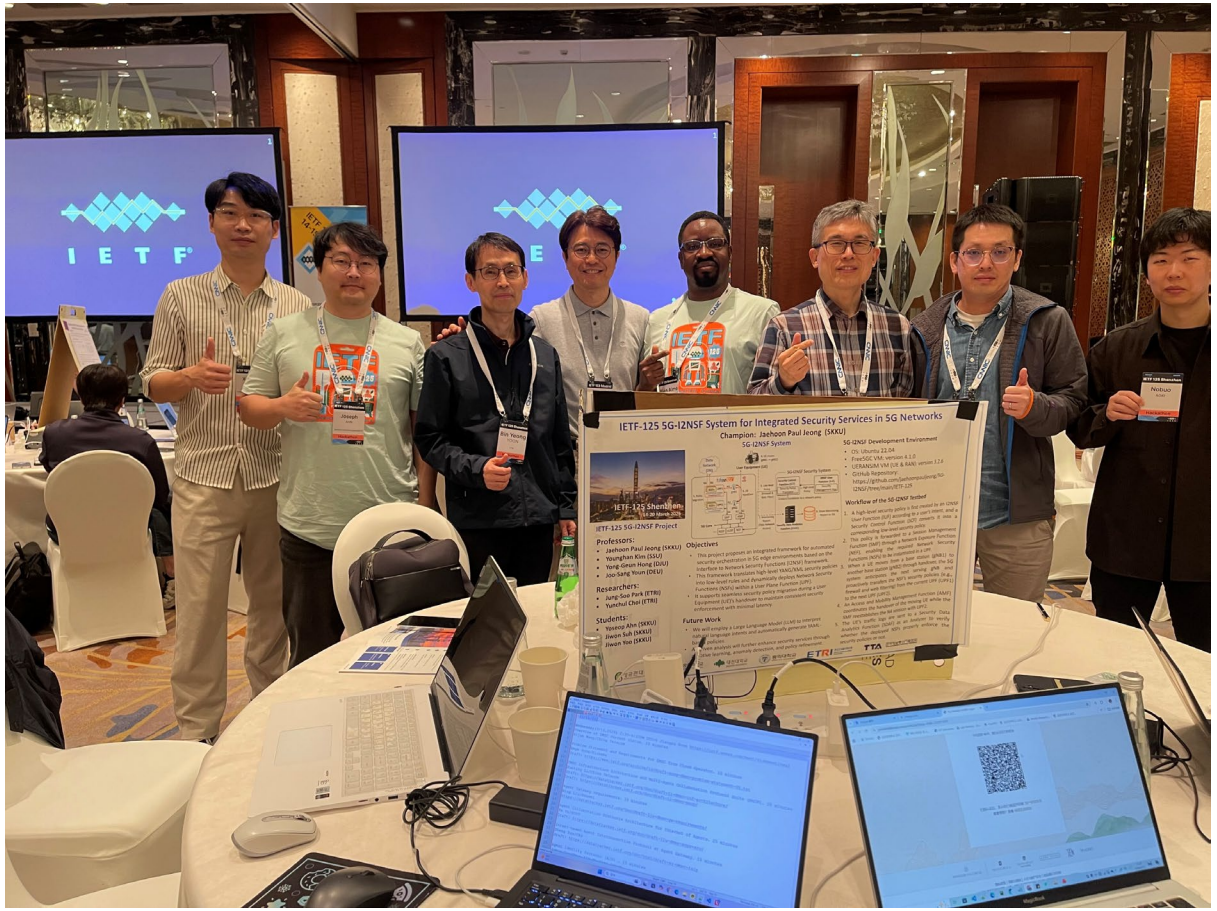
- Jiwon Yoo (SKKU)
- Nobuo Aoki (Independent)

## Hackathon Team Photo